

ANDREA TAVERNITI

SOFTWARE ENGINEER | FULL-STACK, CLOUD & IOT

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Software Engineer experienced in designing and delivering full-stack, cloud and IoT products. Within an R&D environment, I contribute to EU-funded projects through technical analysis, architecture decisions, development, hardware/software integration and technical documentation. I also independently designed and built SNALS Digital Desk, a live Flutter/Firebase platform used by real users.

PROFESSIONAL EXPERIENCE

Software Analyst & Developer | TEA S.R.L.

Jul 2025 – Present

Catanzaro, Calabria — hybrid/remote from Monasterace

- Analyse requirements, design solutions and select technologies for R&D applications and EU-funded projects.
- Deliver full-stack development, evolutionary maintenance and problem solving across existing architectures, while supporting deadlines, technical reviews and project deliverables.
- Coordinate technical activities with universities, company partners and hardware suppliers; prepare reports, presentations and project documentation.
- Integrate software with ESP32 boards, proximity sensors, relays and actuators for multisensory systems and IoT demonstrators.
- Support multispectral acquisition activities on historical parchments in museum environments as part of applied cultural-heritage research.

Freelance Software Developer | Independent and client projects

2019 – Present

Remote

- Independently designed and developed SNALS Digital Desk, a cross-platform Flutter/Firebase platform currently live and used by real users.
- Built more than 80 custom FiveM scripts, including virtual economies, inventories, missions, administration dashboards and React interfaces.
- Managed and optimised MySQL databases containing more than 300,000 records, using asynchronous synchronisation through oxmysql.
- Collaborated with more than 100 communities and private servers, providing integration, documentation and post-release support.
- Delivered websites and web applications for clients using React, WordPress and PHP, including booking, e-commerce, analytics and REST integrations.

SELECTED PROJECTS

SNALS Digital Desk — Flutter, Firebase, Node.js, Google APIs

- End-to-end product including role-based access, multi-location booking, case workflows, real-time chat, QR membership cards, notifications and PDF/CSV reporting.
- Sole ownership of cloud architecture, data model, security rules, PWA and deployment; proposed by a regional representative to the national organisation.

AIVES — React, Python, IoT, automation

- Management tool that standardises project-file generation and enables non-technical operators to configure multisensory experiences.
- Analysed and redesigned sensor logic to prevent unintended reactivation and improve journey reliability.

R&D PROJECTS

MITICA — ESP32, relays, sensors and actuators

- Designed and integrated the AIVES technology demonstrator, covering programming, assembly, configuration, debugging and testing.
- Rapidly acquired electronics skills and fabricated physical components through soldering to complete hardware/software integration.

GAIA — Application and European research

- Defined the initial application structure and contributed to games and configurations supporting children on the autism spectrum.
- Prepared the technical report, presentation, deliverables and documentation; coordinated operational work with university, company and hardware partners.

FiveM Engineering — Lua, React, MySQL, oxmysql

- Complex multiplayer server systems, responsive interfaces and live architecture maintenance focused on stability and resource efficiency.
- Optimised queries, caching and asynchronous synchronisation across large databases.

TECHNICAL SKILLS

Frontend	React, TypeScript, JavaScript, Flutter, Dart, HTML5, CSS3, Tailwind CSS, Bootstrap
Backend	Node.js, Python, PHP, Firebase Functions, REST APIs
Cloud	Firebase Auth, Firestore, Storage, Cloud Functions, Hosting, Cloud Messaging
Databases	MySQL, PostgreSQL, MariaDB, SQLite, MongoDB, Firestore, oxmysql
IoT & Hardware	ESP32, Arduino, proximity sensors, relays, actuators, hardware/software integration
Languages	JavaScript, TypeScript, Python, Dart, Lua, Java, C#, C++, SQL
Tools	Git, GitHub, Docker, VS Code, Firebase, WordPress

EDUCATION

BSc in Computer Engineering University of Calabria	Expected: October 2026
Rende, Italy	
Scientific High School Diploma — 84/100 Pietro Mazzini High School	2011 – 2016
Roccella Jonica, Italy	

LANGUAGES & ADDITIONAL INFORMATION

Italian: native | English: B1 | Driving licence: B